

Differential expression analysis

COURSE ON SCRNA-SEQ DATA ANALYSIS

Test for expression perturbation in single cell data

Jan Kubovčíak · 7.-10. September 2026

with acknowledgments to Luciano Cascione and SIB

Institute of Molecular Genetics of the Czech Academy of Sciences



DEA in single cell

- „Is there significant change in gene expression in different biological conditions?“
- Usage in SC
 - Identify markers of cell clusters/groups
 - Identify transcriptionally perturbed genes between groups of cells from biological replicates

DEA in single cell

1. Identify markers of cell clusters/groups

- Identify genes that are differentially expressed between cell populations
 - e.g., cell types, clusters, or conditions while accounting for the single-cell nature of the data (e.g., sparsity, dropout events)

2. Identify transcriptionally perturbed genes between groups of cells from biological replicates

- perform bulk-like DEA (between groups of samples)

1. Marker identification

- Use cases
 - Annotating cell types (e.g., identifying CD3 as a marker for T cells)
 - Discovering novel cell states or subpopulations
- Methods
 - Pairwise tests (t-test, wilcoxon), linear models etc.
- Challenges
 - Marker genes may not be unique to a single cell type, requiring careful validation
 - Dropout events can obscure marker gene detection

RNA_snn_res.0.1

umap_2

umap_1

0 1 2 3 4 5 6 7

RNA_snn_res.0.3

percent expression

0.25 0.50 0.75 1.00

relative expression

2 1 0 -1

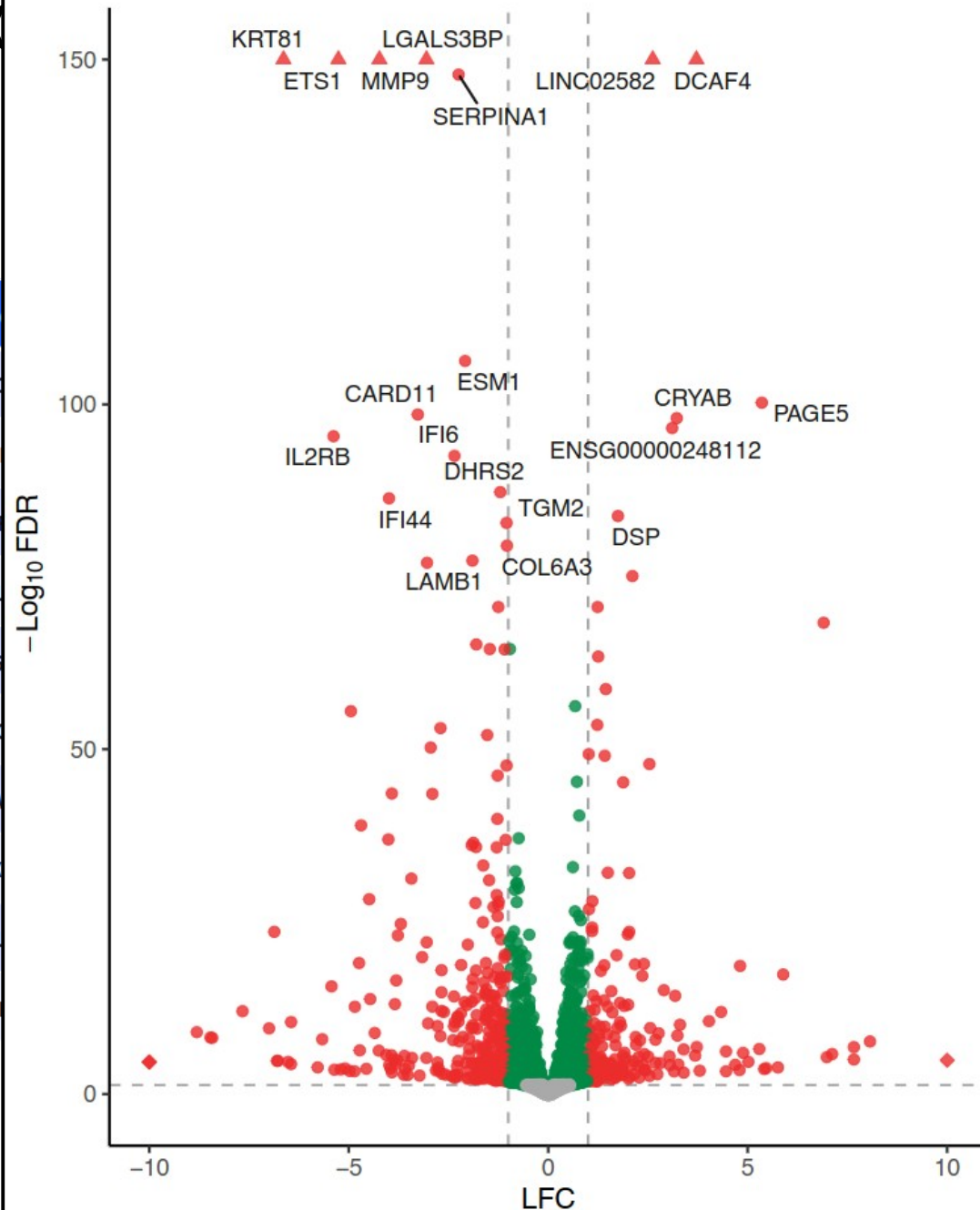
CCR7, BCL11B, CD8B, IFT1, IFT2, IFT3, IFT4, IFT5, IFT6, IFT7, IFT8, IFT9, IFT10, IFT11, IFT12, IFT13, IFT14, IFT15, IFT16, IFT17, IFT18, IFT19, IFT20, IFT21, IFT22, IFT23, IFT24, IFT25, IFT26, IFT27, IFT28, IFT29, IFT30, IFT31, IFT32, IFT33, IFT34, IFT35, IFT36, IFT37, IFT38, IFT39, IFT40, IFT41, IFT42, IFT43, IFT44, IFT45, IFT46, IFT47, IFT48, IFT49, IFT50, IFT51, IFT52, IFT53, IFT54, IFT55, IFT56, IFT57, IFT58, IFT59, IFT60, IFT61, IFT62, IFT63, IFT64, IFT65, IFT66, IFT67, IFT68, IFT69, IFT70, IFT71, IFT72, IFT73, IFT74, IFT75, IFT76, IFT77, IFT78, IFT79, IFT80, IFT81, IFT82, IFT83, IFT84, IFT85, IFT86, IFT87, IFT88, IFT89, IFT90, IFT91, IFT92, IFT93, IFT94, IFT95, IFT96, IFT97, IFT98, IFT99, IFT100, IFT101, IFT102, IFT103, IFT104, IFT105, IFT106, IFT107, IFT108, IFT109, IFT110, IFT111, IFT112, IFT113, IFT114, IFT115, IFT116, IFT117, IFT118, IFT119, IFT120, IFT121, IFT122, IFT123, IFT124, IFT125, IFT126, IFT127, IFT128, IFT129, IFT130, IFT131, IFT132, IFT133, IFT134, IFT135, IFT136, 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What is a marker gene?

Marker id

- Typical out

Volcano plot for contrast 'Rescue_vs_Parental'
FDR ≤ 0.05 , ILFCI > 1 , IF on, shrunken LFC, n = 15822



DEG threshold passed

- LFC and FDR (507)
- FDR only (2371)
- None (12944)

Outliers

- ▲ Above Y limit (6)
- ◆ Below X limit (2)
- ◆ Above X limit (1)

Marker identification

- Special DEA methods for

Analysis | Published: 26 February 2018

Bias, robustness and scalability in single-cell differential expression analysis

[Charlotte Soneson](#) & [Mark D Robinson](#)

[Nature Methods](#) 15, 255–261 (2018) | [Cite this article](#)

59k Accesses | 737 Citations | 139 Altmetric | [Metrics](#)

Abstract

Many methods have been used to determine differential gene expression from single-cell RNA (scRNA)-seq data. We evaluated 36 approaches using experimental and synthetic data and found considerable differences in the number and characteristics of the genes that are called differentially expressed. Prefiltering of lowly expressed genes has important effects, particularly for some of the methods developed for bulk RNA-seq data analysis. However, we found that bulk RNA-seq analysis methods do not generally perform worse than those developed specifically for scRNA-seq. We also present conquer, a repository of consistently processed, analysis-ready public scRNA-seq data sets that is aimed at simplifying method evaluation and reanalysis of published results. Each data set provides abundance estimates for both genes and transcripts, as well as quality control and exploratory analysis reports.

<https://www.nature.com/articles/nmeth.4612>

Marker identification

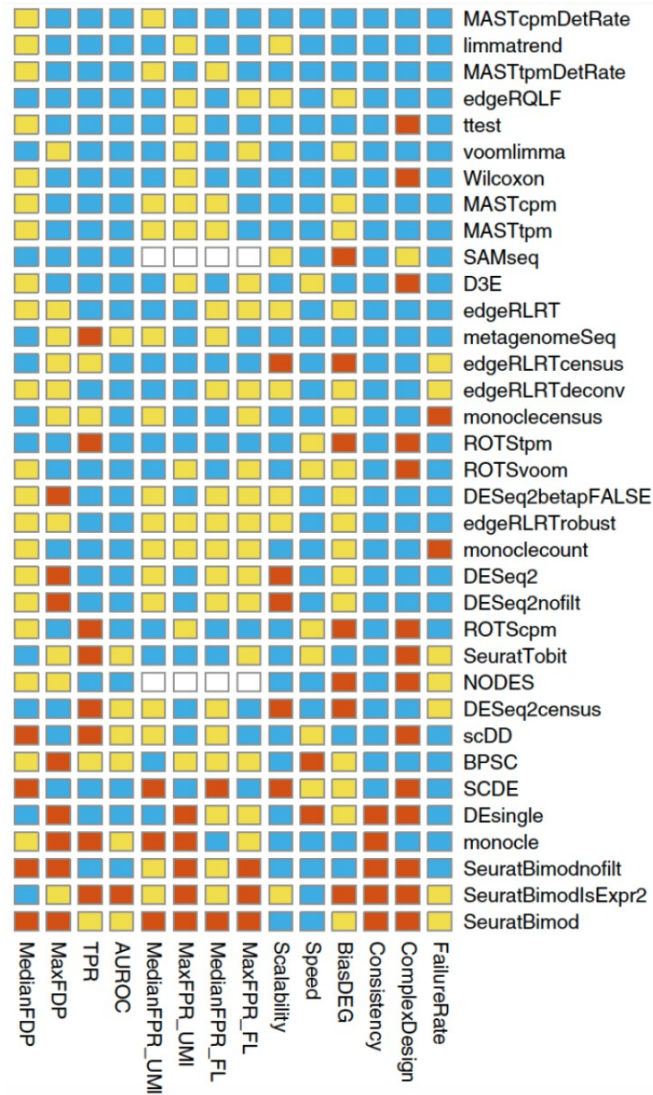


Figure 5 | Summary of DE method performance across all major evaluation criteria. Criteria and cutoff values for performance categories are available in the Online Methods. Methods are ranked by their average performance across the criteria, with the numerical encoding good = 2, intermediate = 1, poor = 0.

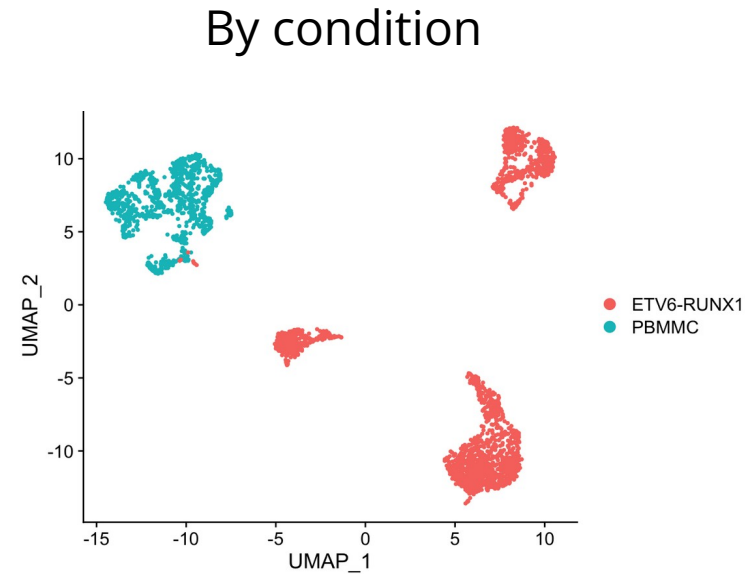
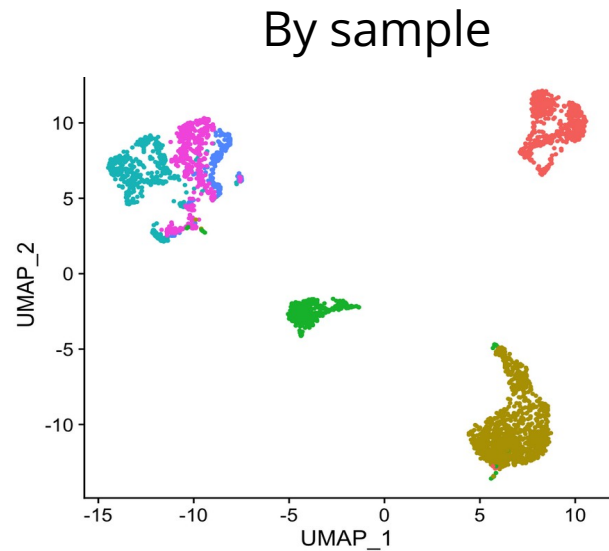
2. Pseudo-bulk DEA

- Use cases
 - Comparing **sample groups**, e.g. treated vs untreated, controls vs patients
 - Complex designs
- Methods
 - Aggregate gene expression counts within groups (e.g., by cell type and sample) and use bulk RNA-seq tools like DESeq2, edgeR, or limma
- Advantages
 - Reduces noise and dropout effects
 - Leverages well-validated bulk RNA-seq tools
- Limitations
 - Loses single-cell resolution and cannot detect cell-to-cell variability

Pseudobulk DEA

- Aggregation

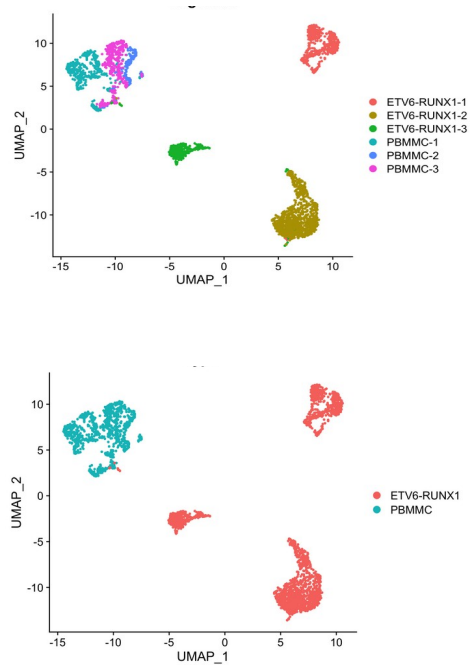
proB cells subset of BMMCs from multiple samples (not integrated)



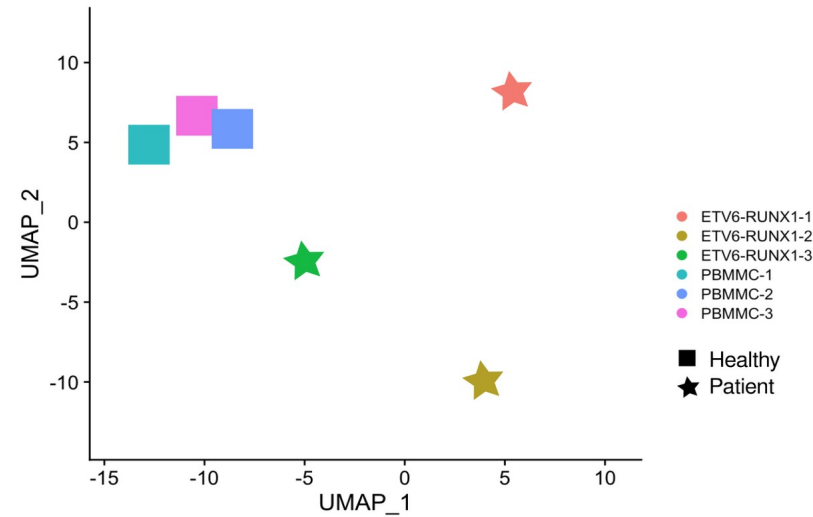
Question: Identify DE genes between healthy and leukemic proB cells

Pseudobulk DEA

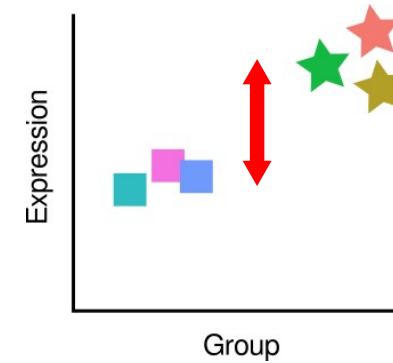
- Aggregation



Pseudobulk by sample & condition



Gene XY

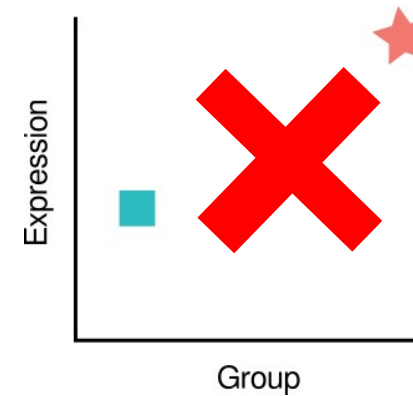
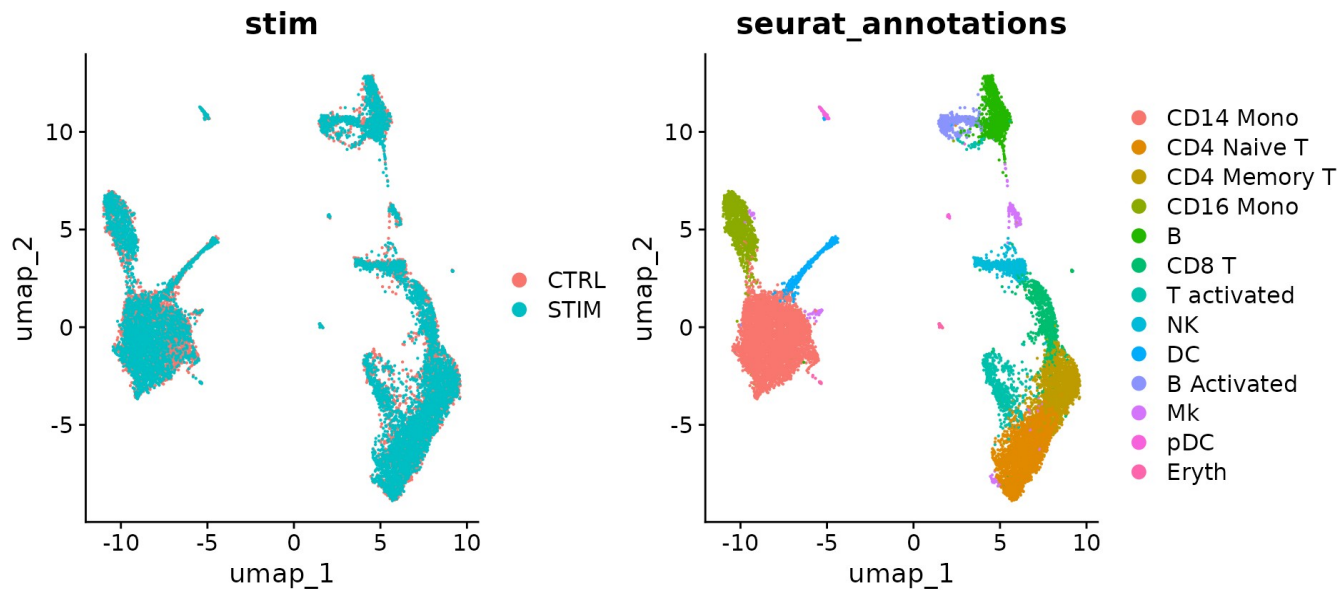


TEST

Pseudoreplication issue

- Pseudo-bulk requires **independent** biological replicates

Interferon-stimulated and control PBMCs, one sample per condition



Thank you for your attention!